

SIMATS ENGINEERING

ASSESSMENT TOOL 2

DSA0202 - COMPUTER VISION WITH OPENCV

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CO2 – ASSESSMENT TOOL-2

Scenario-Based

Code of Conduct:

I Navadeep R (192424110) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Navadeep R

1. Crack Detection on Concrete Surfaces Using Drone Images

Scenario: A civil inspection team is developing a computer vision system to detect thin

cracks on concrete bridges and building surfaces. Images are captured using a drone

under outdoor conditions. Due to changing sunlight, shadows, and camera vibration,

the raw images contain uneven illumination, low contrast, and Gaussian noise.

Questions:

A. Propose a suitable image preprocessing pipeline to prepare the drone images for

crack detection. Your answer should include methods for illumination correction, noise

reduction, contrast enhancement, and image normalization. Justify the need for each

step.

B. Select an appropriate edge detection technique for identifying thin surface cracks.

Compare it with at least one other edge detection method and explain why your

selected technique is more suitable for detecting fine cracks.

C. After edge detection, some crack segments appear broken or disconnected.

Suggest a suitable post-processing technique to connect the crack regions and improve the final detection output.

Scenario

A civil inspection team uses drone images to detect thin cracks on bridges and building surfaces. The images are affected by **uneven illumination, shadows, low contrast, Gaussian noise, and camera vibration**.

A. Suitable Image Preprocessing Pipeline

A suitable preprocessing pipeline is:

Drone Image Acquisition



Illumination Correction



Gaussian Noise Reduction



Contrast Enhancement



Image Normalization



Edge Detection



Morphological Post-processing



Crack Detection

1. Illumination Correction

Outdoor drone images may have bright and dark regions because of sunlight and shadows.

A suitable method is **background estimation followed by illumination correction.**

For example, a blurred version of the image can be considered as the illumination background:

$$I_{corrected} = \frac{I}{B + \epsilon}$$

where:

- I = original image
- B = estimated illumination/background

- ε = small value to avoid division by zero

Purpose: Reduces the effect of shadows and uneven sunlight.

2. Noise Reduction

Since the scenario specifies **Gaussian noise**, a **Gaussian filter** can be used.

The Gaussian filter smooths random intensity variations while maintaining the overall structure.

Purpose:

- Reduce Gaussian noise.
- Prevent noise from being detected as cracks.
- Improve edge detection reliability.

A small kernel such as **3×3 or 5×5** can be used.

3. Contrast Enhancement

Cracks can have very small intensity differences compared with the concrete surface.

CLAHE (Contrast Limited Adaptive Histogram Equalization) is suitable.

CLAHE improves local contrast in different regions while limiting excessive amplification of noise.

Purpose:

- Make thin cracks more visible.

- Improve local intensity differences.
- Handle regions with different illumination.

4. Image Normalization

The enhanced image can be normalized to a fixed intensity range such as:

$$0 - 255$$

or:

$$0 - 1$$

For example:

$$I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}}$$

Purpose:

- Provides consistent intensity values.
- Makes images captured under different lighting conditions more comparable.
- Improves the performance of subsequent algorithms.

Recommended Pipeline

Illumination Correction → Gaussian Filtering → CLAHE → Normalization

This prepares the image for reliable crack detection.

B. Suitable Edge Detection Technique

The **Canny edge detector** is a suitable choice for detecting thin cracks.

Why Canny?

Canny performs several steps:

1. Gaussian smoothing
2. Gradient calculation
3. Non-maximum suppression
4. Double thresholding
5. Edge tracking by hysteresis

This produces thin and well-localized edges.

Canny vs Sobel

Feature	Canny	Sobel
Noise handling	High	Moderate
Edge localization	Excellent	Moderate
Thin crack detection	Very good	Good

Feature	Canny	Sobel
Edge thickness	Thin	Can be thicker
False edges	Lower with proper thresholds	Higher
Computational complexity	Higher	Lower

For fine concrete cracks, **Canny is preferred** because it provides thin, well-localized edges and can suppress weak noise-related edges.

Therefore:

Canny Edge Detector

is a suitable choice.

C. Connecting Broken Crack Segments

After edge detection, cracks may appear broken because of noise, shadows, or weak intensity differences.

A suitable post-processing method is **morphological closing**.

Morphological closing consists of:

Dilation → Erosion

A small structuring element can connect nearby crack segments.

Process

Canny Edge Detection



Dilation



Connect Small Gaps



Erosion



Preserve Crack Structure

Therefore, morphological closing can improve the continuity of detected cracks.

2. Human Motion Tracking in a Security Camera Video

Scenario: A security camera is installed in a hallway to monitor people moving in

different directions. The system must estimate the motion of each person across video

frames for tracking and activity analysis. The hallway contains plain walls, smooth

floors, and occasional shadows.

Questions:

A. Explain how the Lucas-Kanade optical flow method can be used to estimate the

motion of people between consecutive video frames.

B. Discuss how the aperture effect can create errors in motion estimation, especially

on uniform regions such as plain walls, floors, and clothing surfaces.

C. If a person suddenly changes speed or direction, describe how the optical flow

vectors will be affected. Suggest a method to improve tracking accuracy in such

situations.

Scenario

A security camera monitors people moving through a hallway. The system must estimate the motion of people between consecutive frames. The environment contains plain walls, smooth floors, and occasional shadows.

A. Lucas-Kanade Optical Flow

Lucas-Kanade optical flow estimates how image points move between consecutive frames.

It is based on the assumption that the brightness of a moving point remains approximately constant.

The optical flow constraint equation is:

$$I_x u + I_y v + I_t = 0$$

where:

- I_x = intensity change in x-direction
- I_y = intensity change in y-direction
- I_t = intensity change over time
- u = horizontal motion
- v = vertical motion

Since one equation has two unknowns, Lucas-Kanade assumes that nearby pixels have approximately the same motion.

For a person:

Frame 1



Detect features on person



Frame 2



Compare feature positions



Calculate (u,v)



Motion Vector

The resulting vectors indicate:

- Direction of movement
- Approximate speed
- Movement between consecutive frames

For example, if a person moves toward the right:

$$u > 0$$

and the optical flow vector points toward the right.

B. Aperture Effect

The **aperture effect** occurs when only part of an object's motion is visible through a small local region.

Consider a person wearing a shirt with a uniform surface.

If there is little or no texture, the algorithm cannot determine the exact direction of movement.

For example, on a plain wall:

Uniform Wall

No distinctive features



Motion direction difficult to determine



Unreliable optical flow

Similarly, smooth floors or clothing surfaces may provide insufficient visual information.

Effect

The aperture effect can cause:

- Ambiguous motion direction.
- Incorrect motion vectors.

- Unstable tracking.
- Difficulty tracking objects in uniform regions.

Therefore, Lucas-Kanade works best around **corners and textured features** rather than completely uniform regions.

C. Sudden Change in Speed or Direction

If a person suddenly changes speed or direction, the optical flow vectors will also change.

The direction and magnitude of the vectors change.

A sudden speed increase causes the magnitude of the flow vectors to increase.

A sudden direction change causes the vector orientation to change.

Improvement Method

A suitable solution is to combine optical flow with a **Kalman filter**.

The Kalman filter predicts the person's next position based on previous motion and then corrects the prediction using new observations.

Optical Flow



Current Position & Velocity



Kalman Filter



Predict Next Position



New Frame Observation



Update Tracking

This improves tracking stability during sudden changes in speed or direction.

3. Vehicle Tracking at a Traffic Junction Using Video Frames

Scenario: A smart traffic monitoring system uses CCTV footage to track vehicles at a

busy road junction. Vehicles move at different speeds, some stop suddenly at signals,

and others turn left or right. The video also contains shadows, reflections, and partial

occlusion when vehicles overlap.

Questions:

A. Explain how optical flow can be used to estimate the direction and speed of moving

vehicles in the video sequence.

B. Identify two challenges that may affect accurate motion estimation in this scenario,

such as occlusion, shadows, sudden braking, or camera vibration.

Explain how each

challenge affects the optical flow result.

C. Suggest a suitable improvement method to make vehicle tracking more reliable

when vehicles overlap or change direction suddenly.

Scenario

A CCTV system tracks vehicles at a busy junction. Vehicles have different speeds and directions. The scene contains shadows, reflections, sudden braking, turning vehicles, and partial occlusion.

A. Using Optical Flow for Vehicle Tracking

Optical flow estimates the movement of pixels between consecutive video frames.

For each vehicle, the algorithm calculates motion vectors:

$$(u, v)$$

where:

- u = horizontal movement
- v = vertical movement

The magnitude of the motion vector is:

$$M = \sqrt{u^2 + v^2}$$

A larger magnitude generally indicates faster apparent image motion.

The direction is:

$$\theta = \tan^{-1} \left(\frac{v}{u} \right)$$

Example

If the flow vectors are:

→ → → → →

the vehicle is moving toward the right.

If they are:

↑ ↑ ↑ ↑ ↑

the vehicle is moving upward.

If they are:

↗ ↗ ↗ ↗

the vehicle is moving diagonally.

Therefore, optical flow can estimate:

- Vehicle movement direction
- Relative speed
- Turning movement
- Stopping and starting behavior

B. Two Challenges Affecting Motion Estimation

Challenge 1: Occlusion

Occlusion occurs when one vehicle partially or completely covers another vehicle.

Example:

Vehicle A: 

Vehicle B: 



Partial overlap

The optical flow algorithm may not be able to correctly identify which pixels belong to which vehicle.

Effect:

- Incorrect motion vectors.
- Vehicle identity may be lost.
- Tracking may switch between vehicles.
- Motion estimation becomes unreliable.

Challenge 2: Shadows and Reflections

Vehicles can produce shadows and reflections on the road.

The shadows and reflections may move or change even though they are not actual vehicles.

For example:

Actual Vehicle →→→→

Shadow →→→→

The optical flow algorithm may interpret shadow movement as object movement.

Effect:

- False motion vectors.
- Incorrect vehicle boundaries.
- Incorrect speed estimation.
- False vehicle detections.

C. Improvement for Overlap and Sudden Direction Changes

A suitable improvement is to combine **optical flow with object detection and Kalman-filter-based tracking**.

A possible system is:

CCTV Video



Object Detection



Vehicle Identification



Optical Flow



Motion Estimation



Kalman Filter



Vehicle Tracking



Updated Position & Direction

Optical Flow

Provides detailed pixel-level motion information.

Object Detection

Identifies individual vehicles and provides their bounding boxes.

Kalman Filter

Predicts the vehicle's next position when:

- Vehicles overlap.
- A vehicle temporarily disappears.
- Speed changes suddenly.
- Direction changes.

For more complex traffic scenes, a **multi-object tracking algorithm** can also maintain a unique ID for each vehicle.